Supplementary Information

Physics-Aware Lightweight Segmentation for Quantitative Holographic Cell Analysis with Optical Mass Preservation

Usama Ikram[1], Youhyun Kim[2], and Inkyu Moon[1,2,*]

[1]Department of Artificial Intelligence, DGIST, Daegu 42988, South Korea

[2]Department of Robotics & Mechatronics Engineering, DGIST, Daegu 42988, South Korea

[*]Corresponding author: inkyu.moon@dgist.ac.kr (I. Moon)

S-I. LoRA Rank Sweep for All Architectures

Table S-1 lists the segmentation and integrated-phase results of the full objective for every architecture and LoRA rank (r = 2, 4, 8, 16, and 32), and Fig. S-1 shows every training seed. The main text uses EdgeSAM (r=32) as the primary configuration and MobileSAM (r=8) as the sensitivity comparison (shaded rows); this choice is descriptive and not the result of a selection rule. The dependence on rank was not monotonic for any architecture. Training was unstable in some seeds (Fig. S-2): EdgeSAM lost the spherocyte class in seed 44 at r = 2, 8, and 16, MobileSAM (r=32) collapsed in seed 43 in the second epoch and lost two classes, and MobileNet-UNet lost at least one class in every seed at every rank. Latency and memory were measured for the rank-8 models only (Table 4 of the main text).

TABLE S-1 LoRA rank sweep, full objective. Mean ± SD over seeds 42, 43, and 44; shaded: configurations used in the main text.

Architecture	LoRA rank	Trainable params	Mean Dice	AJI	Boundary F1	Global IPE (%)	Seeds with a lost class (mask values)
EdgeSAM	2	0.30 M (5.2%)	0.625 ± 0.108	0.249 ± 0.017	0.598 ± 0.020	4.45 ± 2.71	1/3 (seed 44: 2)
	4	0.38 M (6.5%)	0.703 ± 0.009	0.252 ± 0.021	0.615 ± 0.012	3.48 ± 0.91	0/3
	8	0.53 M (8.9%)	0.647 ± 0.114	0.249 ± 0.043	0.608 ± 0.009	3.81 ± 1.82	1/3 (seed 44: 2)
	16	0.85 M (13.4%)	0.665 ± 0.114	0.277 ± 0.022	0.623 ± 0.024	4.66 ± 3.53	1/3 (seed 44: 2)
	32	1.47 M (21.2%)	0.738 ± 0.006	0.335 ± 0.020	0.642 ± 0.005	3.02 ± 0.84	0/3
MobileSAM	2	0.63 M (9.4%)	0.808 ± 0.003	0.264 ± 0.013	0.801 ± 0.006	1.30 ± 0.09	0/3
	4	0.64 M (9.6%)	0.811 ± 0.004	0.268 ± 0.037	0.803 ± 0.002	1.34 ± 0.07	0/3
	8	0.67 M (10.0%)	0.810 ± 0.006	0.262 ± 0.017	0.802 ± 0.011	1.32 ± 0.09	0/3
	16	0.73 M (10.8%)	0.813 ± 0.002	0.279 ± 0.054	0.801 ± 0.013	1.29 ± 0.18	0/3
	32	0.85 M (12.3%)	0.687 ± 0.234	0.283 ± 0.020	0.794 ± 0.023	1.76 ± 0.53	1/3 (seed 43: 2, 4)
MobileNet-UNet	2	5.47 M (71.4%)	0.410 ± 0.004	0.534 ± 0.029	0.816 ± 0.006	2.61 ± 0.23	3/3 (seed 42: 2, 4; seed 43: 2, 4; seed 44: 2, 4)
	4	5.51 M (71.6%)	0.406 ± 0.002	0.560 ± 0.035	0.805 ± 0.006	2.50 ± 0.19	3/3 (seed 42: 2, 4; seed 43: 2, 4; seed 44: 2, 4)
	8	5.58 M (71.8%)	0.412 ± 0.004	0.551 ± 0.053	0.814 ± 0.003	2.60 ± 0.07	3/3 (seed 42: 2, 4; seed 43: 2, 4; seed 44: 2, 4)
	16	5.73 M (72.4%)	0.414 ± 0.002	0.505 ± 0.012	0.805 ± 0.009	3.09 ± 0.65	3/3 (seed 42: 2, 4; seed 43: 2, 4; seed 44: 2, 4)
	32	6.04 M (73.4%)	0.476 ± 0.108	0.541 ± 0.049	0.824 ± 0.014	5.83 ± 5.75	3/3 (seed 42: 2, 4; seed 43: 2, 4; seed 44: 2)

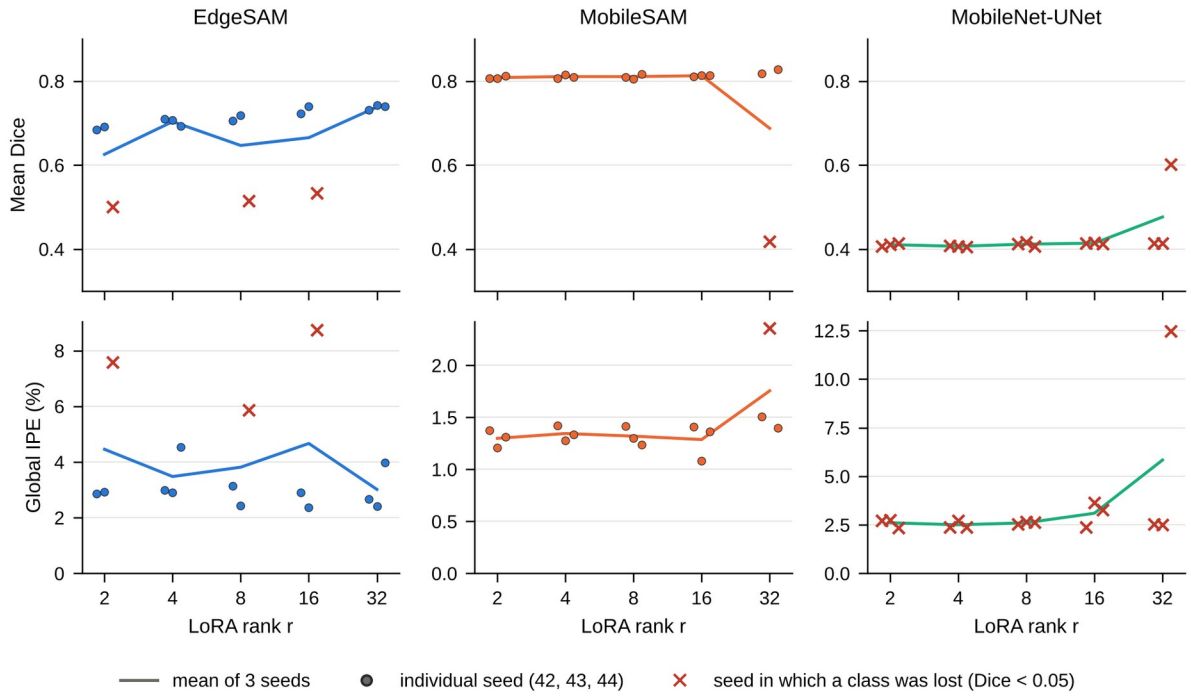


Fig. S-1 Mean Dice (top) and global IPE (bottom) against LoRA rank, full objective. Dots: individual seeds; crosses: seeds in which a class was lost (Dice < 0.05); line: mean of three seeds.

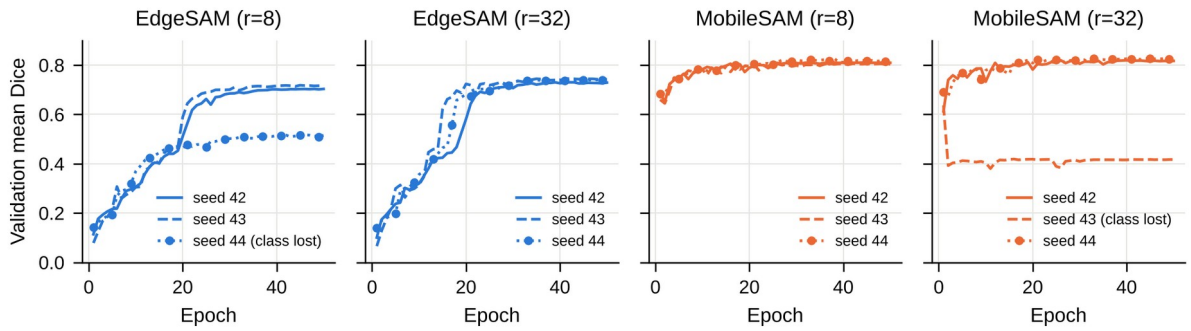


Fig. S-2 Validation mean Dice during training of EdgeSAM and MobileSAM at r=8 and r=32, one line per seed; seeds that ended with a lost class are marked in the legend.

S-II. Loss Ablation at All LoRA Ranks

Table S-2 gives the component-wise ablation (D: Dice only; P: Dice with PMC; B: Dice with PMC and BGA; F: full objective with IPP) for every architecture and rank. Over the 15 architecture–rank configurations, the mean global IPE of the full objective was lower than that of Dice alone in 15 and lower than that of Dice with PMC and BGA in 13; the seed-paired 95% interval excluded zero in 3 and 1 configurations, respectively. Over the 45 seed-paired comparisons, the full objective had the lower IPE in 42 against Dice alone and in 36 against Dice with PMC and BGA. A class was lost in 6, 3, and 13 of the 15 Dice-only runs of EdgeSAM, MobileSAM, and MobileNet-UNet, and in 3, 1, and 15 of the 15 full-objective runs. For EdgeSAM, the AJI of the full objective was higher than that of Dice with PMC and BGA in 0 of 15 seed-paired comparisons, and its Dice was higher than with Dice alone in 9 of 15.

TABLE S-2 Loss ablation at all LoRA ranks (means over seeds 42, 43, and 44).

Architecture	LoRA rank	Mean Dice (D / P / B / F)	Global IPE, % (D / P / B / F)	Seeds with a lost class (D / P / B / F)
EdgeSAM	2	0.576 / 0.710 / 0.717 / 0.625	14.42 / 5.26 / 4.88 / 4.45	0 / 0 / 0 / 1
	4	0.610 / 0.727 / 0.723 / 0.703	10.96 / 4.68 / 4.98 / 3.48	1 / 0 / 0 / 0
	8	0.624 / 0.672 / 0.735 / 0.647	10.82 / 7.97 / 4.25 / 3.81	1 / 1 / 0 / 1
	16	0.621 / 0.741 / 0.742 / 0.665	11.05 / 3.71 / 3.43 / 4.66	2 / 0 / 0 / 1

Architecture	LoRA rank	Mean Dice (D / P / B / F)	Global IPE, % (D / P / B / F)	Seeds with a lost class (D / P / B / F)
	32	0.619 / 0.680 / 0.746 / 0.738	11.20 / 7.45 / 3.64 / 3.02	2 / 1 / 0 / 0
MobileSAM	2	0.741 / 0.805 / 0.740 / 0.808	5.35 / 1.55 / 5.32 / 1.30	1 / 0 / 1 / 0
	4	0.811 / 0.806 / 0.741 / 0.811	1.44 / 1.50 / 5.23 / 1.34	0 / 0 / 1 / 0
	8	0.811 / 0.813 / 0.813 / 0.810	1.53 / 1.60 / 1.48 / 1.32	0 / 0 / 0 / 0
	16	0.813 / 0.813 / 0.817 / 0.813	1.54 / 1.58 / 1.55 / 1.29	0 / 0 / 0 / 0
	32	0.674 / 0.817 / 0.818 / 0.687	9.23 / 1.58 / 1.46 / 1.76	2 / 0 / 0 / 1
MobileNet-UNet	2	0.416 / 0.413 / 0.549 / 0.410	17.98 / 19.88 / 12.40 / 2.61	3 / 3 / 3 / 3
	4	0.482 / 0.417 / 0.490 / 0.406	16.52 / 19.39 / 14.70 / 2.50	3 / 3 / 3 / 3
	8	0.485 / 0.414 / 0.544 / 0.412	15.48 / 19.79 / 13.96 / 2.60	2 / 3 / 2 / 3
	16	0.550 / 0.413 / 0.612 / 0.414	14.18 / 20.13 / 11.50 / 3.09	2 / 3 / 2 / 3
	32	0.359 / 0.486 / 0.418 / 0.476	20.83 / 15.96 / 19.54 / 5.83	3 / 3 / 3 / 3

S-III. Class-Specific Integrated-Phase Error and Dice

Table S-3 lists the class-specific IPE and Dice of the configurations of Table 1 and of the EdgeSAM loss ablation. A class that is not predicted in an image gives an IPE of 100% for that image, so the class-specific IPE is dominated by misclassification between morphologies; the global IPE over the whole foreground is the measure of integrated-phase preservation used in the main text.

TABLE S-3 Class-specific IPE (%) and Dice by mask value (1 discocyte, 2 spherocyte, 3 echinocyte, 4 stomatocyte), means over seeds 42, 43, and 44, including seeds in which a class was lost.

Configuration	Loss	IPE (%), 1	IPE (%), 2	IPE (%), 3	IPE (%), 4	Dice, 1	Dice, 2	Dice, 3	Dice, 4
MobileNet-UNet (*r*=8)	+ PMC + BGA + IPP (full)	79.5	100.0	28.2	100.0	0.891	0.000	0.756	0.000
MobileSAM (*r*=8)	+ PMC + BGA + IPP (full)	7.2	22.1	28.7	55.2	0.923	0.851	0.760	0.708
EdgeSAM (*r*=8)	+ PMC + BGA + IPP (full)	22.5	51.5	34.3	49.6	0.882	0.513	0.628	0.563
EdgeSAM (*r*=32)	+ PMC + BGA + IPP (full)	6.4	24.2	28.7	52.2	0.895	0.789	0.655	0.614
EdgeSAM (*r*=8)	Dice	6.5	64.2	28.7	49.4	0.893	0.343	0.668	0.592
EdgeSAM (*r*=8)	+ PMC	7.4	49.5	29.7	47.1	0.892	0.524	0.669	0.601
EdgeSAM (*r*=8)	+ PMC + BGA	7.6	22.1	31.5	46.7	0.892	0.783	0.654	0.612
EdgeSAM (*r*=32)	Dice	7.0	71.2	30.8	43.6	0.898	0.272	0.675	0.630
EdgeSAM (*r*=32)	+ PMC	6.5	48.4	28.0	47.8	0.896	0.522	0.678	0.625
EdgeSAM (*r*=32)	+ PMC + BGA	6.4	24.8	30.4	44.6	0.896	0.781	0.678	0.627

S-IV. Segmentation Accuracy versus Inference Throughput

Fig. S-3 relates the mean Dice of the rank-8 models to their FP32 forward-pass throughput (Table 4 of the main text); both quantities refer to rank 8. Dice was evaluated with the PyTorch checkpoints and is plotted at the same value for both runtimes.

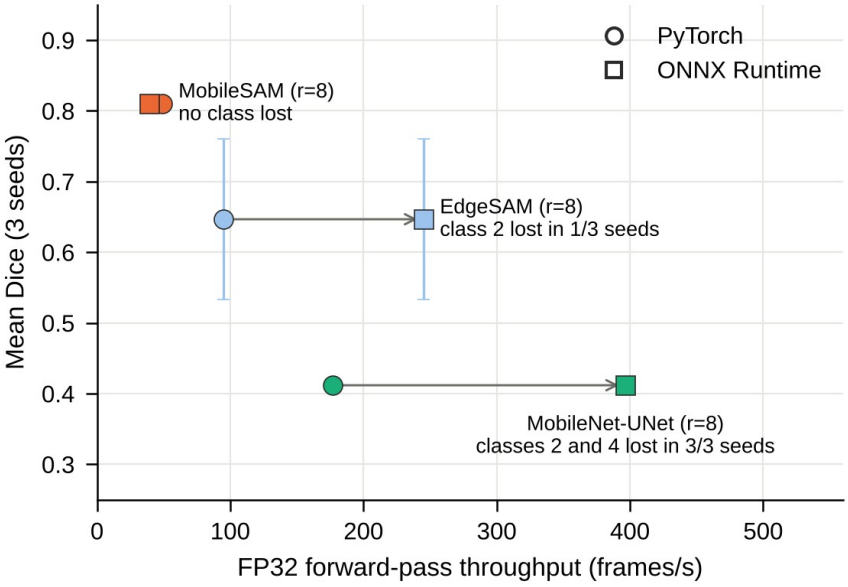


Fig. S-3 Mean Dice (mean ± SD over three seeds) against FP32 forward-pass throughput (mean of three timing passes) of the rank-8 models. Circles: PyTorch; squares: ONNX Runtime; class losses are noted next to each model.

S-V. Phase-Gradient Alignment of Predicted Contours

Fig. S-4 shows predicted cell contours on the phase-gradient magnitude for one validation window, as a qualitative illustration of the boundary cue used by the BGA term. Its quantitative effect is assessed by the loss ablation (Table 3 of the main text and Table S-2).

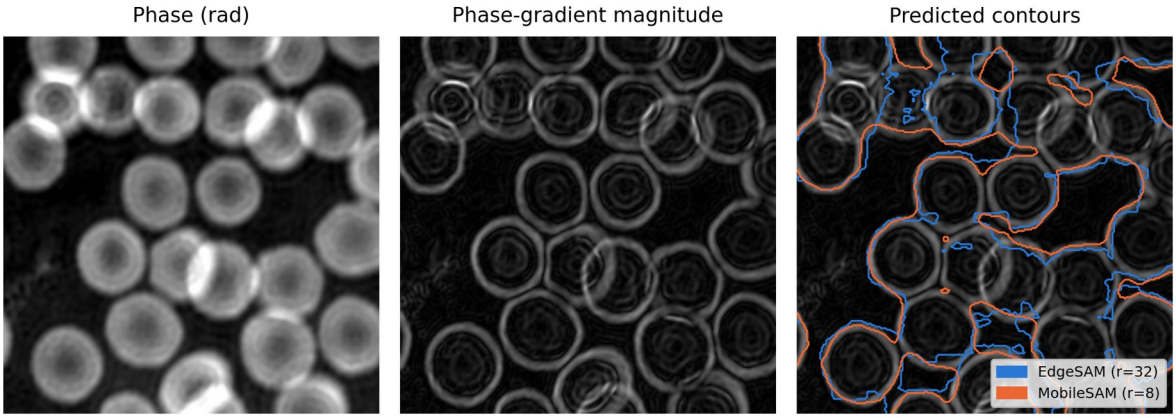


Fig. S-4 Phase map, phase-gradient magnitude, and contours of EdgeSAM (r=32) and MobileSAM (r=8) (seed 42) on the gradient magnitude. The window is the 256 × 256 px window with the most annotated foreground on a 32 px grid within the validation image with the most annotated foreground; it coincides with the discocyte window of Fig. 6 because discocytes dominate that image.

S-VI. Representative Phase Images over the Storage Period

Fig. S-5 shows one validation phase map from each of five storage days, with the EdgeSAM (r=32) contours.

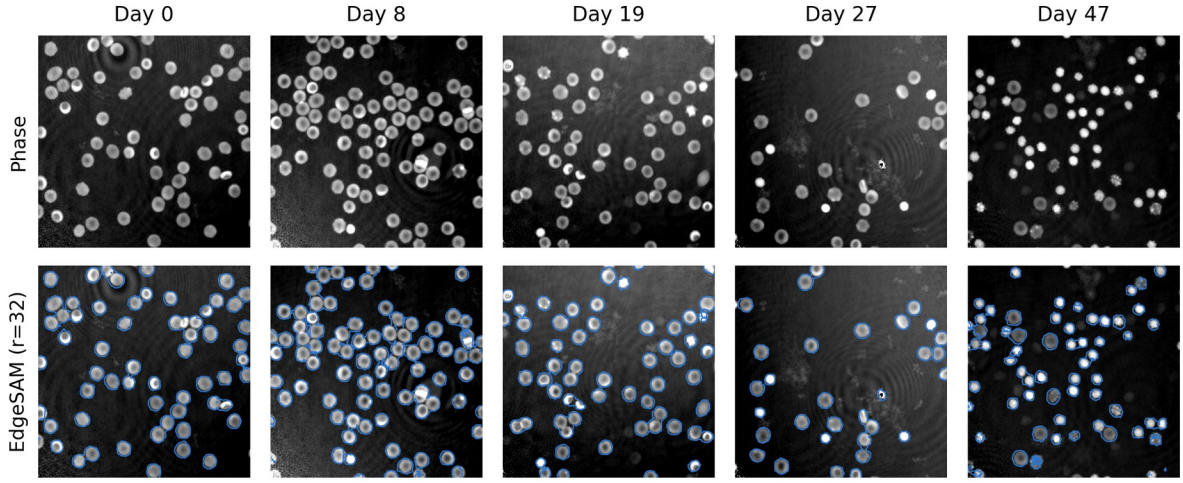


Fig. S-5 Phase maps from storage days 0, 8, 19, 27, and 47 (first validation image of each day); bottom row: EdgeSAM (*r*=32) contours (seed 42).

S-VII. Storage-Period Measurements per Day

Tables S-4 and S-5 give the per-day measurements behind Fig. 8 for EdgeSAM (*r*=32) and MobileSAM (*r*=8). Cells are 4-connected foreground components of 15 to 200 µm² that do not touch an unannotated region; phase-wrapped cells are excluded. Annotation columns give the mean and, in parentheses, the cell-level SD; prediction columns give the mean over seeds of the per-day cell mean and, in brackets, the smallest and largest seed value. Predicted components are not cell counts. On days marked †, fewer than 0.6 kept components were found per annotated cell, and the predicted means describe only the components that passed the filters. Fig. S-6 shows the kept-component ratio and the fate of all predicted components per day.

TABLE S-4 Per-day measurements: annotation and EdgeSAM (*r*=32), seeds 42, 43, and 44.

Day	Annotated cells	Area, annotation (µm²)	Dry mass, annotation (pg)	Kept components (seeds 42 / 43 / 44)	Kept / annotated	Area, prediction (µm²)	Dry mass, prediction (pg)
0	185	52.1 (21.6)	31.7 (14.3)	142 / 146 / 154	0.80	50.5 [48.4–52.6]	31.5 [30.5–32.2]
5	211	51.2 (21.9)	33.2 (15.0)	139 / 144 / 159	0.70	58.5 [56.8–60.2]	37.3 [36.7–37.7]
8	220	51.9 (24.4)	34.9 (17.8)	119 / 114 / 112	0.52 †	59.3 [55.9–64.9]	39.3 [37.4–42.3]
12	230	54.9 (31.8)	36.3 (22.3)	103 / 108 / 124	0.49 †	61.6 [58.6–64.0]	40.2 [38.6–41.2]
15	206	59.3 (32.6)	40.9 (25.1)	51 / 53 / 74	0.29 †	60.7 [59.6–61.7]	41.6 [40.1–42.8]
19	162	49.8 (23.7)	32.2 (15.6)	124 / 124 / 130	0.78	54.5 [50.7–57.7]	35.5 [33.6–36.8]
23	108	49.5 (20.3)	30.6 (12.4)	92 / 95 / 95	0.87	49.8 [48.4–50.7]	31.3 [30.8–31.7]
27	93	50.8 (22.9)	33.0 (13.3)	84 / 85 / 89	0.92	49.9 [48.6–51.5]	33.2 [32.7–34.4]
30	188	46.0 (24.0)	33.5 (15.2)	160 / 164 / 161	0.86	44.5 [43.9–44.8]	32.8 [32.5–33.0]
37	157	36.7 (16.8)	31.0 (10.4)	144 / 153 / 156	0.96	35.5 [33.9–37.3]	30.1 [28.9–31.2]
47	228	27.1 (11.0)	28.8 (7.5)	180 / 192 / 183	0.81	28.9 [28.2–29.3]	30.1 [29.4–30.6]

TABLE S-5 Per-day measurements: annotation and MobileSAM (*r*=8), seeds 42, 43, and 44.

Day	Annotated cells	Area, annotation (µm²)	Dry mass, annotation (pg)	Kept components (seeds 42 / 43 / 44)	Kept / annotated	Area, prediction (µm²)	Dry mass, prediction (pg)
0	185	52.1 (21.6)	31.7 (14.3)	138 / 138 / 140	0.75	52.5 [52.0–53.3]	33.2 [32.9–33.8]
5	211	51.2 (21.9)	33.2 (15.0)	142 / 150 / 145	0.69	57.1 [56.1–58.4]	37.7 [37.2–38.5]
8	220	51.9 (24.4)	34.9 (17.8)	104 / 105 / 107	0.48 †	55.3 [54.8–55.8]	37.3 [36.9–37.6]
12	230	54.9 (31.8)	36.3 (22.3)	105 / 104 / 106	0.46 †	59.7 [58.4–61.0]	39.7 [39.0–40.6]
15	206	59.3 (32.6)	40.9 (25.1)	34 / 32 / 40	0.17 †	56.8 [51.2–62.7]	39.1 [35.7–42.8]
19	162	49.8 (23.7)	32.2 (15.6)	127 / 127 / 130	0.79	52.2 [51.0–54.0]	34.7 [34.0–35.5]
23	108	49.5 (20.3)	30.6 (12.4)	89 / 91 / 93	0.84	50.4 [49.4–51.6]	31.7 [31.3–32.1]
27	93	50.8 (22.9)	33.0 (13.3)	81 / 81 / 83	0.88	52.9 [52.6–53.2]	35.5 [35.2–35.7]
30	188	46.0 (24.0)	33.5 (15.2)	156 / 146 / 150	0.80	47.4 [45.9–48.2]	35.4 [34.8–35.8]
37	157	36.7 (16.8)	31.0 (10.4)	147 / 146 / 146	0.93	39.2 [38.8–39.6]	32.8 [32.4–33.2]
47	228	27.1 (11.0)	28.8 (7.5)	205 / 193 / 197	0.87	29.5 [28.0–30.6]	31.5 [30.5–32.3]

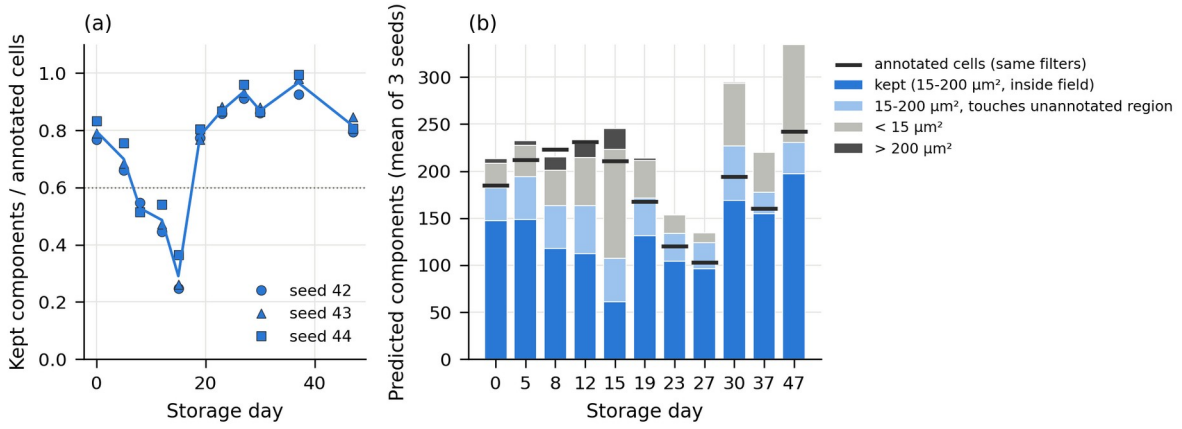


Fig. S-6 Predicted components of EdgeSAM (*r*=32) per storage day: (a) kept components divided by annotated cells (markers: seeds); (b) fate of the predicted components inside the annotated field (mean of three seeds, wrapped components included), with the annotated cells kept by the same filters.

S-VIII. Date-Clustered Storage Statistics

Table S-6 gives, for each segmentation and training seed, the image-level Spearman correlation with storage day, the permutation p-value with the 11 acquisition dates permuted as whole blocks (exact test over all 11! = 39,916,800 permutations, so the p-values carry no Monte Carlo error), the 95% date-bootstrap interval (20,000 resamples), and the range of the correlation when each date is left out, for projected area, dry mass, and the per-image difference between predicted and annotation-derived dry mass. With 11 clusters and an undocumented donor structure, these tests are approximate and exploratory. For comparison, the table also gives the p-value of the image-level Spearman test that treats the 66 images as independent. That test is anti-conservative here, because the six images of a date share the same samples: for the EdgeSAM (*r*=32) dry mass it gives p = 0.008, 0.052, 0.036 for seeds 42, 43, and 44, whereas the date-clustered test gives p = 0.10, 0.17, 0.090. Conclusions in the main text are based on the date-clustered test.

TABLE S-6 Date-clustered storage statistics of the per-image medians (wrapped cells excluded).

Segmentation	Seed	Quantity (per-image median)	Spearman ρ	p (66 images as independent)	p (date permutation)	95% date-bootstrap interval	Leave-one-date-out range
Annotation	–	Area	−0.53	< 0.0001	0.018	−0.85 to 0.04	−0.64 to −0.38
Annotation	–	Dry mass	−0.11	0.36	0.27	−0.33 to 0.11	−0.20 to −0.04
EdgeSAM (*r*=32)	42	Area	−0.75	< 0.0001	0.0008	−0.93 to −0.27	−0.84 to −0.66
EdgeSAM (*r*=32)	43	Area	−0.60	< 0.0001	0.007	−0.87 to −0.10	−0.68 to −0.48
EdgeSAM (*r*=32)	44	Area	−0.60	< 0.0001	0.005	−0.86 to −0.10	−0.68 to −0.47
EdgeSAM (*r*=32)	42	Dry mass	−0.32	0.008	0.10	−0.56 to 0.07	−0.49 to −0.27
EdgeSAM (*r*=32)	43	Dry mass	−0.24	0.052	0.17	−0.47 to 0.06	−0.35 to −0.18
EdgeSAM (*r*=32)	44	Dry mass	−0.26	0.036	0.090	−0.48 to 0.05	−0.41 to −0.20
EdgeSAM (*r*=32)	42	Dry-mass difference (prediction − annotation)	−0.29	0.020	0.080	−0.46 to 0.05	−0.39 to −0.22
EdgeSAM (*r*=32)	43	Dry-mass difference (prediction − annotation)	−0.21	0.085	0.20	−0.43 to 0.07	−0.31 to −0.15
EdgeSAM (*r*=32)	44	Dry-mass difference (prediction − annotation)	−0.23	0.059	0.088	−0.41 to 0.03	−0.34 to −0.18
MobileSAM (*r*=8)	42	Area	−0.59	< 0.0001	0.005	−0.87 to −0.08	−0.66 to −0.46
MobileSAM (*r*=8)	43	Area	−0.59	< 0.0001	0.006	−0.86 to −0.10	−0.67 to −0.46
MobileSAM (*r*=8)	44	Area	−0.49	< 0.0001	0.024	−0.81 to 0.04	−0.60 to −0.33
MobileSAM (*r*=8)	42	Dry mass	−0.14	0.27	0.35	−0.38 to 0.15	−0.27 to −0.08
MobileSAM (*r*=8)	43	Dry mass	−0.20	0.11	0.24	−0.45 to 0.15	−0.36 to −0.15
MobileSAM (*r*=8)	44	Dry mass	−0.16	0.20	0.34	−0.43 to 0.18	−0.32 to −0.09
MobileSAM (*r*=8)	42	Dry-mass difference (prediction − annotation)	−0.03	0.80	0.78	−0.18 to 0.14	−0.11 to 0.01
MobileSAM (*r*=8)	43	Dry-mass difference (prediction − annotation)	−0.05	0.69	0.63	−0.23 to 0.14	−0.14 to −0.008

Segmentation	Seed	Quantity (per-image median)	Spearman ρ	p (66 images as independent)	p (date permutation)	95% date-bootstrap interval	Leave-one-date-out range
MobileSAM (*r*=8)	44	Dry-mass difference (prediction − annotation)	−0.003	0.98	0.98	−0.22 to 0.22	−0.11 to 0.04

S-IX. Per-Seed Storage Measurements

Table S-7 lists, for EdgeSAM (*r*=32) and MobileSAM (*r*=8), every training seed and storage day: the number of kept components, their ratio to the annotated cells, the mean and cell-level SD of projected area and dry mass, the mean circularity, and the share of components assigned to each class (mask values 1 to 4: discocyte, spherocyte, echinocyte, stomatocyte). Cells are not pooled across seeds; phase-wrapped cells are excluded. The annotation values for each day are given in Tables S-4 and S-5.

TABLE S-7 Per-seed, per-day measurements of the kept predicted components (six images per day). Kept: number of kept components; Kept / ann.: ratio to the annotated cells of the day; area and dry mass: mean (cell-level SD); Class 1 to 4: share of components by mask value.

Configuration	Seed	Day	Kept	Kept / ann.	Area (µm²)	Dry mass (pg)	Circularity	Class 1 (%)	Class 2 (%)	Class 3 (%)	Class 4 (%)
EdgeSAM (*r*=32)	42	0	142	0.77	52.6 (22.5)	32.2 (13.9)	0.854	93.0	0.0	2.1	4.9
		5	139	0.66	60.2 (29.7)	37.7 (18.4)	0.810	95.0	0.0	0.7	4.3
		8	119	0.54	64.9 (38.6)	42.3 (26.6)	0.764	93.3	0.0	1.7	5.0
		12	103	0.45	64.0 (36.8)	41.2 (23.9)	0.762	84.5	1.0	5.8	8.7
		15	51	0.25	61.7 (34.3)	41.8 (23.3)	0.682	86.3	0.0	5.9	7.8
		19	124	0.77	57.7 (33.1)	36.8 (19.9)	0.792	83.1	2.4	7.3	7.3
		23	92	0.85	50.7 (20.7)	31.7 (12.5)	0.839	82.6	3.3	8.7	5.4
		27	84	0.90	49.6 (26.0)	32.7 (14.0)	0.841	70.2	10.7	14.3	4.8
		30	160	0.85	44.8 (25.6)	33.0 (14.7)	0.809	54.4	21.9	19.4	4.4
		37	144	0.92	37.3 (19.6)	31.2 (12.9)	0.831	34.7	45.1	18.1	2.1
		47	180	0.79	29.3 (14.5)	30.2 (10.7)	0.808	11.1	66.7	21.1	1.1
	43	0	146	0.79	50.4 (24.2)	31.8 (14.8)	0.875	91.8	0.0	2.1	6.2
		5	144	0.68	58.4 (30.3)	37.5 (18.8)	0.814	92.4	0.0	1.4	6.2
		8	114	0.52	57.1 (30.8)	38.1 (21.1)	0.789	87.7	0.0	4.4	7.9
		12	108	0.47	58.6 (34.7)	38.6 (22.9)	0.786	83.3	0.9	7.4	8.3
		15	53	0.26	59.6 (40.2)	40.1 (23.7)	0.736	75.5	5.7	5.7	13.2
		19	124	0.77	55.2 (33.0)	36.1 (19.6)	0.795	75.0	3.2	8.1	13.7
		23	95	0.88	50.2 (23.7)	31.5 (13.4)	0.837	84.2	4.2	9.5	2.1
		27	85	0.91	51.5 (26.8)	34.4 (14.8)	0.841	71.8	10.6	14.1	3.5
		30	164	0.87	44.8 (26.7)	33.0 (15.3)	0.796	54.3	21.3	22.0	2.4
		37	153	0.97	35.2 (16.5)	30.1 (10.7)	0.857	34.6	45.1	18.3	2.0
		47	192	0.84	29.2 (13.8)	30.6 (11.8)	0.827	10.9	65.6	22.4	1.0
	44	0	154	0.83	48.4 (20.3)	30.5 (12.8)	0.869	90.3	0.0	1.9	7.8
		5	159	0.75	56.8 (29.0)	36.7 (17.8)	0.809	89.9	0.0	1.9	8.2
		8	112	0.51	55.9 (32.7)	37.4 (22.1)	0.775	89.3	0.0	3.6	7.1
		12	124	0.54	62.2 (37.6)	40.9 (24.5)	0.746	83.1	0.8	4.8	11.3
		15	74	0.36	60.7 (38.3)	42.8 (27.0)	0.678	82.4	4.1	8.1	5.4
		19	130	0.80	50.7 (27.5)	33.6 (17.7)	0.804	76.9	2.3	6.2	14.6
		23	95	0.88	48.4 (19.9)	30.8 (12.2)	0.827	78.9	4.2	11.6	5.3
		27	89	0.96	48.6 (22.7)	32.7 (12.9)	0.820	69.7	11.2	15.7	3.4
		30	161	0.86	43.9 (25.0)	32.5 (14.8)	0.795	53.4	20.5	23.0	3.1
		37	156	0.99	33.9 (14.9)	28.9 (8.2)	0.849	34.6	46.2	17.9	1.3
		47	183	0.80	28.2 (12.7)	29.4 (9.5)	0.826	10.9	69.9	19.1	0.0
MobileSAM (*r*=8)	42	0	138	0.75	52.2 (24.3)	33.0 (15.4)	0.920	90.6	0.0	2.9	6.5
		5	142	0.67	58.4 (30.9)	38.5 (20.5)	0.874	92.3	0.0	1.4	6.3
		8	104	0.47	55.8 (28.1)	37.6 (17.8)	0.872	93.3	0.0	3.8	2.9

Configuration	Seed	Day	Kept	Kept / ann.	Area (µm²)	Dry mass (pg)	Circularity	Class 1 (%)	Class 2 (%)	Class 3 (%)	Class 4 (%)
		12	105	0.46	61.0 (37.3)	40.6 (24.4)	0.860	85.7	1.0	3.8	9.5
		15	34	0.17	56.6 (31.2)	38.8 (20.4)	0.862	79.4	2.9	11.8	5.9
		19	127	0.78	54.0 (29.1)	35.5 (18.2)	0.882	77.2	2.4	6.3	14.2
		23	89	0.82	51.6 (20.9)	32.1 (13.1)	0.909	86.5	1.1	10.1	2.2
		27	81	0.87	52.8 (29.4)	35.7 (17.2)	0.895	75.3	9.9	13.6	1.2
		30	156	0.83	45.9 (26.6)	34.8 (16.7)	0.898	54.5	18.6	22.4	4.5
		37	147	0.94	38.8 (21.4)	32.4 (14.7)	0.935	34.7	39.5	23.8	2.0
		47	205	0.90	28.0 (13.2)	30.5 (11.3)	0.931	8.8	61.5	29.3	0.5
	43	0	138	0.75	53.3 (25.3)	33.8 (15.7)	0.910	93.5	0.0	2.2	4.3
		5	150	0.71	56.1 (27.5)	37.2 (18.2)	0.884	93.3	0.0	1.3	5.3
		8	105	0.48	54.8 (25.4)	36.9 (17.3)	0.876	92.4	0.0	3.8	3.8
		12	104	0.45	58.4 (33.7)	39.0 (22.7)	0.872	87.5	1.0	3.8	7.7
		15	32	0.16	62.7 (36.3)	42.8 (24.7)	0.837	81.2	3.1	3.1	12.5
		19	127	0.78	51.6 (26.9)	34.5 (16.8)	0.898	78.0	2.4	6.3	13.4
		23	91	0.84	50.2 (19.6)	31.7 (12.4)	0.916	84.6	2.2	12.1	1.1
		27	81	0.87	52.6 (28.5)	35.2 (16.7)	0.906	75.3	8.6	13.6	2.5
		30	146	0.78	48.2 (27.5)	35.7 (17.3)	0.891	58.2	17.8	19.9	4.1
		37	146	0.93	39.6 (20.9)	33.2 (13.7)	0.920	34.2	41.1	23.3	1.4
		47	193	0.85	30.6 (17.4)	32.3 (14.4)	0.921	9.8	61.1	28.5	0.5
	44	0	140	0.76	52.0 (24.5)	32.9 (15.1)	0.921	92.1	0.0	2.1	5.7
		5	145	0.69	56.9 (28.4)	37.5 (18.8)	0.882	94.5	0.0	1.4	4.1
		8	107	0.49	55.4 (29.3)	37.3 (19.0)	0.877	92.5	0.0	3.7	3.7
		12	106	0.46	59.6 (36.2)	39.6 (23.8)	0.869	84.9	0.9	3.8	10.4
		15	40	0.19	51.2 (24.6)	35.7 (15.3)	0.895	80.0	5.0	7.5	7.5
		19	130	0.80	51.0 (25.7)	34.0 (16.3)	0.900	74.6	1.5	6.9	16.9
		23	93	0.86	49.4 (19.6)	31.3 (12.5)	0.919	83.9	2.2	11.8	2.2
		27	83	0.89	53.2 (28.9)	35.5 (16.8)	0.904	71.1	9.6	15.7	3.6
		30	150	0.80	48.2 (28.7)	35.8 (17.6)	0.896	58.0	19.3	19.3	3.3
		37	146	0.93	39.2 (21.8)	32.7 (14.7)	0.930	34.2	39.0	24.0	2.7
		47	197	0.86	29.8 (15.0)	31.8 (12.9)	0.924	9.6	63.5	26.4	0.5